

VANCA RAFAEL-MARIAN

EMBEDDED SOFTWARE ENGINEER

Indie App Developer



CONTACT

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[LinkedIn Page](#)

[Github Page](#)

[My personal Page](#)

SKILLS

- Project Management
- Simulink / Matlab
- C/C++/ Firmware/ Low - Level
- Dart/ Flutter
- Jira/ Polarion/ Confluence/ Agile
- Git/ Github/ Jenkins/ Github Actions
- SpeedGoat/ DSpace/ Microautobox
- Google AdMob/ Console
- Electronics/ Inverters/ EV's/ FOC
- Signal Processing/ PIC/ Renesas
- Figma Design/ Firebase/ Netlify
- Hardware Design/ electronics

LANGUAGES

- English (Fluent)
- Romanian (Fluent)
- German (Basic)

DRIVING LICENSE

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PROFILE

Embedded Software Engineer focused on Advanced R&D and production within the automotive sector and chemical sector using micro-controllers. Skilled in Matlab, Simulink, C/C++ and Hardware Integration, with a strong commitment to delivering high-quality code in Agile environments. Passionate about development and apps, creating and publishing multiple apps on Google Play.

Take a look at my personal page: <https://vancarafaelfmarian.netlify.app/>

If you wanna contact me and I'm not responding on one of those two phone numbers, please use WhatsApp or email because of low phone signal due to EMC room and metal frame building. Thank you.



WORK EXPERIENCE

Hanna Instruments

2026 - PRESENT

Embedded Software Engineer

- Firmware Development: Engineered and maintained robust firmware in C/C++ for PIC and Renesas micro-controllers, ensuring high reliability for precision analytical instrumentation.
- Low-Level Programming: Interfaced directly with hardware registers and peripherals (I2C, SPI, UART) to optimize memory management, power consumption, and processing speed.
- Acted as the primary technical escalation point for a 40-person production unit, resolving critical hardware and software issues across 12 distinct product lines.
- Engineered integrated hardware-software test stands that allow manufacturing staff to seamlessly mount PCBs, flash micro controller firmware, and run functional diagnostics.

Schaeffler

2024 - 2026

Embedded Software Engineer

- Developed safety-critical control log for Inverters using Matlab/Simulink, focusing on signal scaling, HDL synthesis, and code validation.
- Managed version control and automation using Git/GitHub, Jenkins, and GitHub Actions to streamline CI/CD pipelines.
- Validated software performance on hardware targets (SpeedGoat, MicroAutoBox 2&3) using CANalyzer and oscilloscopes for rigorous signal analysis.
- Produced comprehensive technical documentation and collaborated in an Agile team to transition innovation projects from concept to Business Unit integration.

Digilent

2023- 2023

Student - Embedded Software

- Digilent VHDL club working with FPGA and System-on-Chip based on ARM processor.
- Working with Vitis SDK to develop different project using C Embedded for Zybo Z7 and Basys 3 development boards.
- Develop and implement different tasks on Basys 3 and Zybo Z7 boards using Vivado with VHDL.



EDUCATION

Master of Circuits and Integrated Systems

2023 - 2025

School of Electronics | Technical University of Cluj Napoca

Final grade: 10.0 / 10.0

Bachelor of Electronic Telecommunication and Information Technology

2019 - 2023

Applied Electronics | Technical University of Cluj Napoca

Final grade: 9.4 / 10.0

AI-Optimized Field Oriented Control (FOC) System

[Link to project](#)

- Designed a 3-phase grid-connected power system, integrating an AC/DC rectifier, DC-DC Boost converter, and battery storage simulation.
- Implemented Field Oriented Control (FOC) for a 3-Level inverter, utilizing Reinforcement Learning (RL) to optimize current control loops and system efficiency.
- Established CI/CD pipelines using GitHub Actions to automate testing and ensure continuous integration.

FPGA Sound Equalizer & Visualizer

[Link to project](#)

- Developed a real-time sound equalizer on the Zybo Z7 FPGA, utilizing Vivado (VHDL) for hardware definition and Vitis (Embedded C) for application logic.
- Implemented I2C communication with the SSM2603 Audio Codec and designed a Finite State Machine (FSM) to manage audio recording, processing, and playback.
- Integrated VGA output for a graphical sound bar display and maintained version control via Git/GitHub.

Flutter & Dart Apps for IOS and Google Play more than 10

[Link to project](#)

- Developed a cross-platform (Mobile & Web) application using Flutter and publish them on Google Play Console gaining constant users on all platforms, links on my Personal Page.
- Creating Ads and monetizing the apps with Google market and Google AdMob.
- Creating Web pages for each app in particular and hosting them on Netlify. They are created with React + Vue.
- Fitness, Accounting, Battery Monitor, Budget, Bible, Emf Radar, Focus Productivity, Photo Editor, Light Meter, Decibel Meter, Money tracking, Plant Identification, Glow Up, League of Legends Builds, Pressure Meter, Vibration Analyzer and many more apps to come.

PCB Design Projects

[Link to project](#)

- Designed multi-layer PCB layouts using OrCAD Capture CIS, managing stack-up and component routing.
- Performed electrical validation and circuit testing to ensure design integrity.

8051 Environmental Monitoring System

[Link to project](#)

- Developed an embedded system using the 8051 microcontroller, interfacing analog/digital sensors via ADC and I2C protocols to display data on an LCD.
- Implemented low-level firmware in Assembly using Keil uVision 5 and validated the circuit design through Proteus simulation

Liquid Level Control System

[Link to project](#)

- Designed an analog monitoring circuit in Orcad, utilizing a potentiometer-based sensing mechanism and LED indicators.
- Conducted rigorous circuit stability analysis using DC-Sweep, Transient, Monte Carlo, and Worst-Case simulations to validate performance under varying conditions.

Image Processing – Laplacian & Gaussian

[Link to project](#)

- Implemented Laplacian edge detection algorithms in Python, automating the generation of convolution masks for various dimensions.
- Optimized algorithm performance for high-speed processing and real-time contouring applications suitable for practical deployment.